

FULL TEST - 01

Rough

Ans 1. (3) b and c

Ans 2. (4) All of these

Ans 3. (1) Tallness is dominant over pea plants.

Ans 4. (3) b, c

Ans 5. (3) Translocation

Ans 6. (1) Asteries

Ans 7. (3) a convex mirror

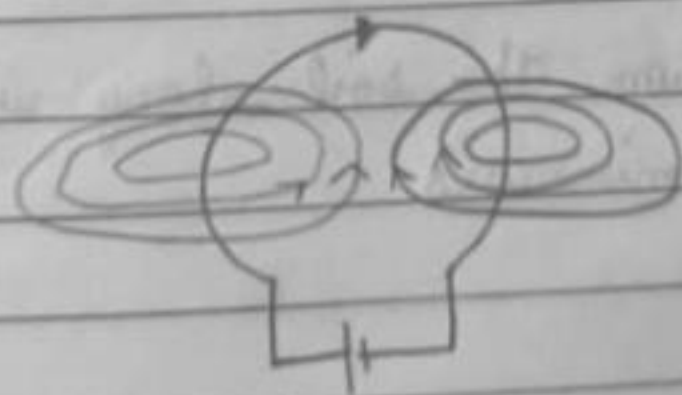
Ans 8. (4)

Ans 9. (1) Scattering of light is not enough at---

Ans 10. (2) Refraction, Dispersion and Total Internal ref---

Ans 11. (1) low resistance and low melting point.

Ans 12. (2)



Ans 13:

Ans 13. (3) The pattern of magnetic field inside a solenoid - - - bar magnet.

Ans 14. (2) endothermic reaction

Ans 15. (1) CO_2

Ans 16. (1) 1

Ans 17. (iv) $2\text{ZnO} (s) \rightarrow$

Ans 17. (3) (i) and (iii)

Ans 18. (3) i - c, ii - d, iii - b, iv - a

Ans 19. (2) $\text{CH}_3 - \text{CH}_2 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} - \underset{\text{CH}_3}{\overset{\text{OH}}{\text{C}}} - \text{CH}_3$

Ans 20. (2) 148

Ans 21. The plants get the raw materials required for photosynthesis from the atmosphere and from the soil.

Plants like Carbon dioxide and oxygen are taken from the atmosphere.

Water is taken up from the soil along with other nutrients and minerals.

Ans 22: Haemoglobin is a red-pigment present in our blood. It helps in transporting oxygen as it has high affinity for oxygen.

If there is deficiency of Haemoglobin in our blood, then it will hamper the transport of blood with oxygen in our body. Oxygen will not be reaching every part of our body and the synthesis of food would stop.

Ans 23: When the angle between the current and the magnetic field is 0° or 180° then the current carrying conductor placed in a magnetic field would not experience any force.

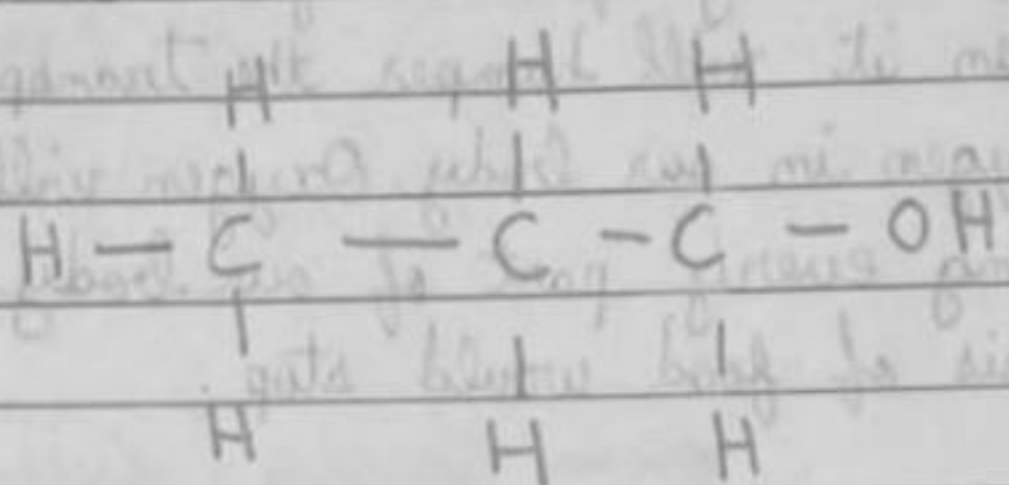
If the magnetic field becomes parallel to the direction of current, no force will be experienced by the conductor.

Ans 24: The three characteristics of the image formed by a plane mirror is:

- ① The image is Erect and Virtual.
- ② The image is of the same size as the object i.e., its magnification is $+1$.
- ③ The image is laterally inverted.

Ans 25. The alcohol with 3 Carbon atoms in its molecule is ~~Ethanol~~ Propanol. Its formula is ~~C_2H_5OH~~ C_3H_7OH

Its structure is:



Propanol $[CH_3CH_2CH_2OH \text{ or } C_3H_7OH]$

Ans 26. (i) Magnesium

(ii) Copper

^{Calcium}
~~Aluminium~~ and Sodium react with cold water.

Aluminium reacts with steam.

Ans 27. Lungs are the organs which help in carrying out the exchange of gases inside the human body.

The lungs are caged inside 12 pair of bones which together make up the ribcage.

When the air is taken in or inhaled, the following changes occur:

- ① The ribcage expands.
- ② The diaphragm stretches and almost becomes straight.
- ③ The volume of thoracic cavity increases.

When we exhale air the following changes happen:

- ① The ribcage contracts.
- ② The diaphragm comes back to its normal shape i.e., curved.
- ③ The volume of thoracic cavity decreases.

The diaphragm is a muscle present at the bottom of the lungs which help in exchange of gases.

Ans 28. (a) Liver has function of releasing and secreting Bile juice. Bile juice helps in emulsification of fats.

Pancreas have a function of releasing pepsin, trypsin and lipase which help in digesting proteins and fats.

(b) Nutrition is a process of intake and utilization of food materials by our body.

Nutrients are the substances that help in

growth and development of an organism and include carbohydrates, fats, proteins etc.

Holozoic Nutrition

- ① The nutrition in which one organism eats the other organism for food.

- ② Ex: Human, tiger etc.

Saprophytic Nutrition

The nutrition in which the organism depends on dead and decaying matter for food.

- Ex: Mushroom, Fungi etc.

Ans 29: Self Pollination:

- ① In, this type of pollination, the pollination occurs in the same plant.

- ② One flower of the plant pollinates another flower of the same plant.

- ③ The pollination may happen either in the same flower or in two different flowers of the same plant.

Cross Pollination:

- ① In, this type of pollination, the pollination occurs in two different plant.

- (2) The flower of one plant pollinates the flower of another plant.

The various agents of pollination are:

- (1) Water (2) Wind
(3) Insects - Honeybees, Butterflies, Birds etc.

Ans 30. The Joules law of heating states that:

When a current passes through a conductor then the heat produced is equal to the product of resistance, time and the square of current flowing.

It can be concluded mathematically that:

$$H = I^2 R T.$$

Also as $V = IR$ (Ohm's Law)

$$\text{So, } H = \frac{V^2}{R} T$$

$$\text{and Also, } H = V I T.$$

$$\text{and as } P = IV$$

$$\text{So, } H = P T.$$

(L) length of wire = 12 m
radius of wire = 2×10^{-4} m

So, Area of cross section of wire (A) = πr^2
 $A = \pi (2 \times 10^{-4})^2$
 $A = 4\pi \times 10^{-8} \text{ m}^2$

and Resistivity (ρ) = $3.14 \times 10^{-8} \Omega \text{ m}$.

Now, $R = \frac{\rho L}{A}$ where R is resistance.

So, $R = \frac{(3.14 \times 10^{-8}) \times 12}{4\pi \times 10^{-8}}$

$$R = \frac{3.14 \times 12 \times 10^{-8}}{4 \times 3.14 \times 10^{-8}} \quad [\text{Taking } \pi = 3.14]$$

$$R = \frac{12}{4}$$

$$R = 3 \Omega$$

So, the Resistance is 3Ω .

Ans 31. (i) For correcting his distant vision:

$$P = -0.5 \text{ D}$$

also, $P = \frac{1}{f}$ where f is focal length in metres.

$$\text{So, } -0.5 = \frac{1}{f}$$

$$f = -\frac{10}{5}$$

$$f = -2 \text{ m or } -200 \text{ cm.}$$

So, for correcting his distant vision he required focal length of $-2 \text{ m or } -200 \text{ cm.}$

(ii) For correcting his near vision:

$$P = +1.5 \text{ D}$$

$$\text{Also, } P = \frac{1}{f}$$

$$\text{So, } 1.5 = \frac{1}{f}$$

$$f = \frac{10}{15} = \frac{2}{3}$$

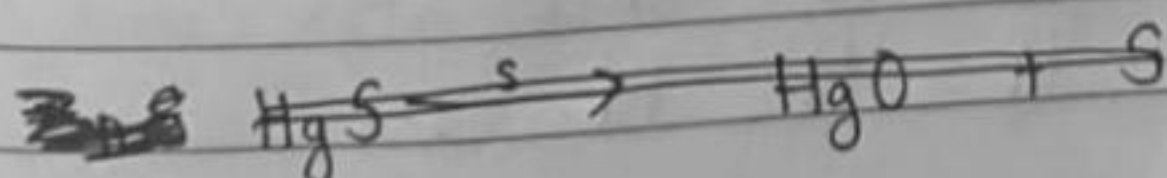
$$f = +0.66 \text{ m}$$

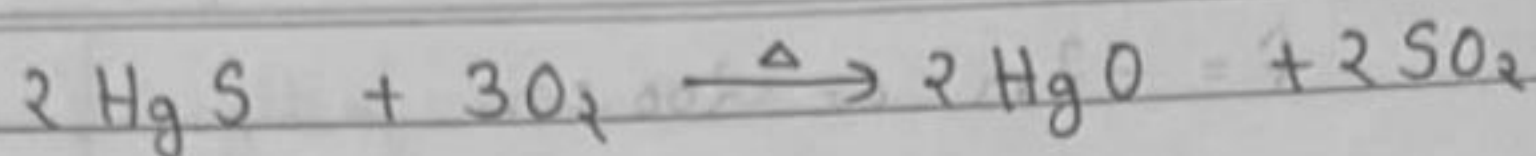
So, for correcting his near vision he requires focal length of $+0.66 \text{ m or } +66.66 \text{ cm.}$

As 3. The Metal is Mercury (Hg).

It is found as a compound with Sulphur called Mercury Sulphide HgS or Linalbar.

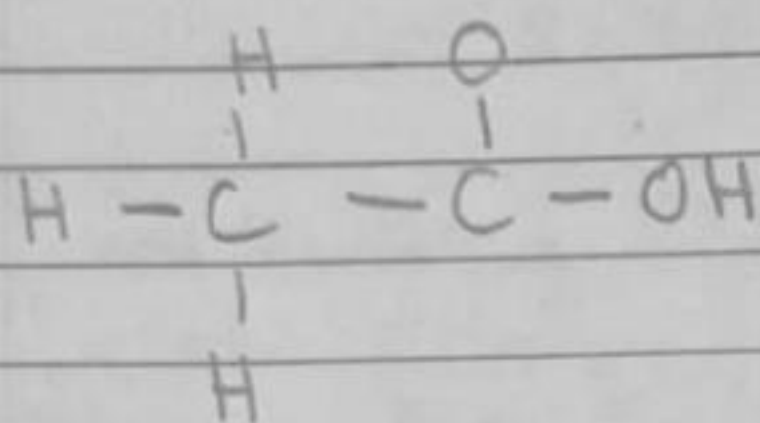
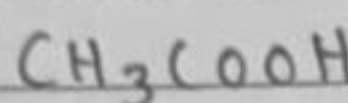
The reaction involved is called roasting in which sulphide ores are strongly heated to obtain oxide ores. It is as follows:





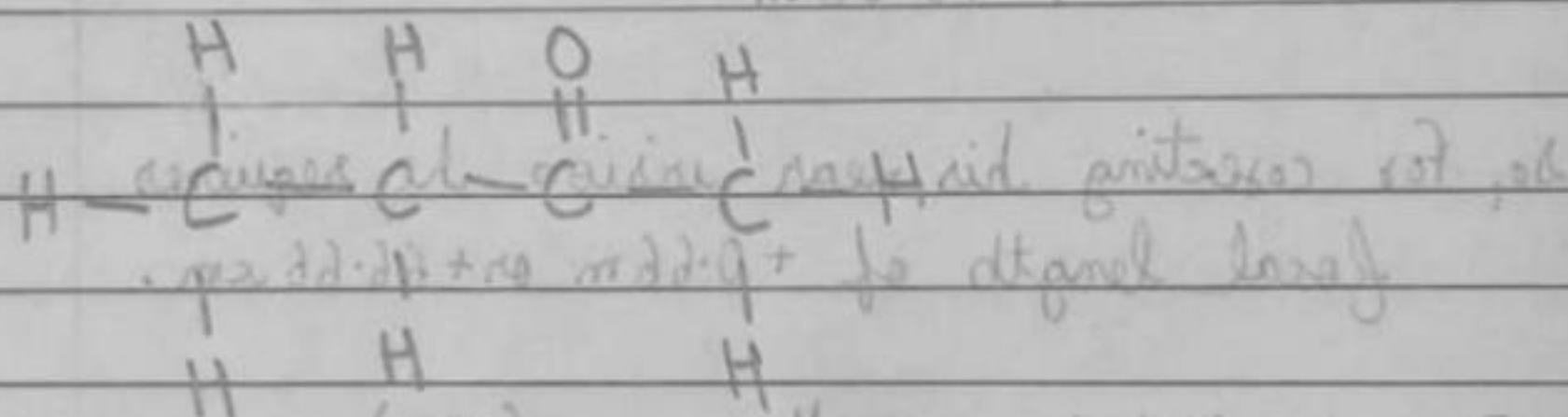
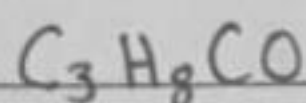
Ans 33. The structures of the compounds are:

(i) Ethanoic Acid:



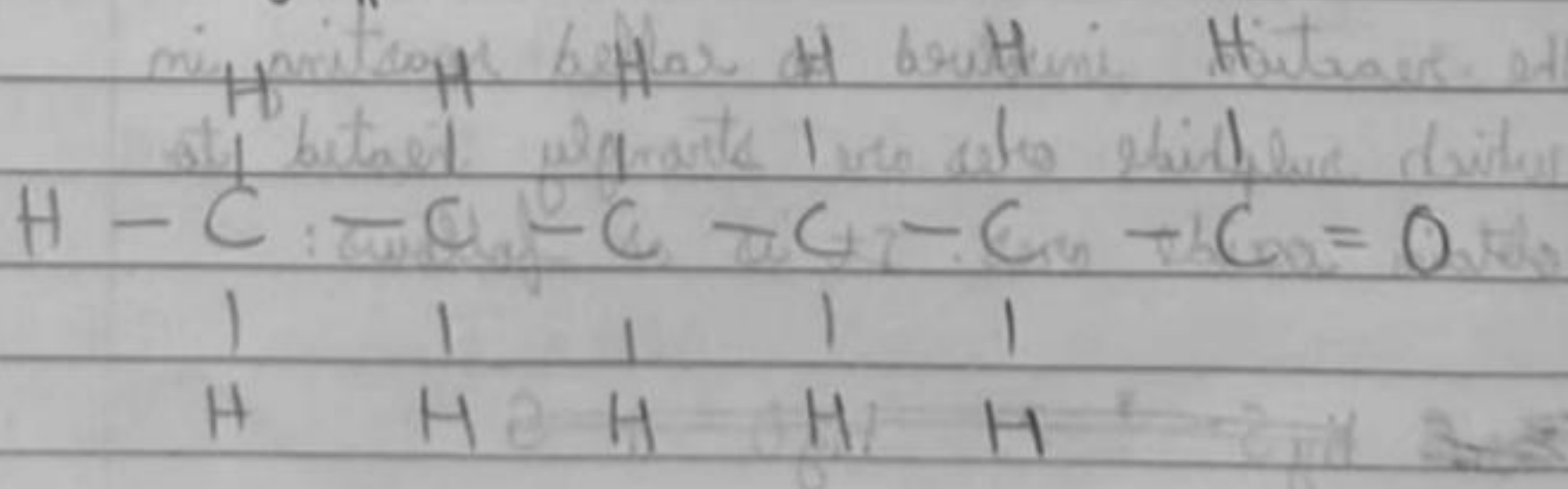
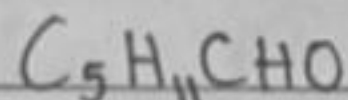
Ethanoic Acid.

(ii) Butanone:



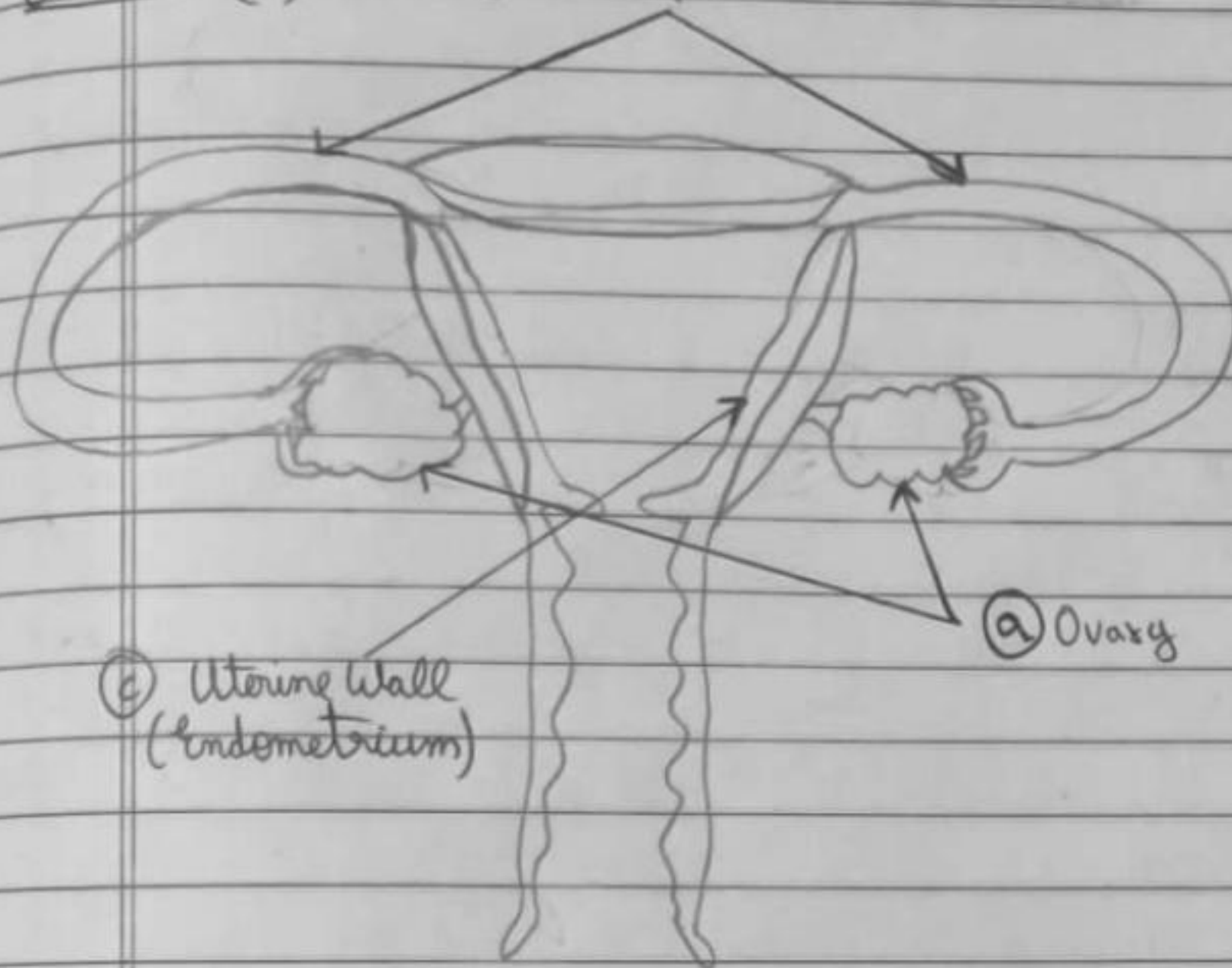
Butan-2-one

(iii) Hexanal:



Ans 34. (i)

(b) Fallopian Tube



FEMALE REPRODUCTIVE ORGAN.

→

a) that produces eggs - ovary

b) where fusion of egg and sperm takes place - Fallopian Tube

c) where zygote is implanted - Uterine wall (Endometrium).

(ii) When an egg is not fertilised then it starts to move down the fallopian tube. After some time it gets released

along with blood and mucus through the vagina. This process is known as menstruation. This occurs every one month when the egg is not fertilized.

Ans 35. (i) When the bulb B₁ gets fused then the current in the other two bulbs will increase. But as the glowing of bulb is not dependent on current, so the other two bulbs glow will remain same.

(ii) The ammeter A₂ will read current of 0 A.

The reading on ammeter A₁ and A₃ will increase.

as the bulbs A₁ and B₂ are identical so their resistance will be equal.

Now, By Ohm's law,

$$\text{Resistance of the entire circuit} = \frac{V}{I}$$

$$R_{\text{net}} = \frac{4.5}{3}$$

$$R_{\text{net}} = 1.5 \Omega.$$

Now as, the potential difference is same and the resistance of the two bulbs are also same, so the current will be same.

As, $I_1 + I_2 = I$ (iii)

So, $I_1 + I_2 = I$

$2 I_1 = 3$

to get $I_1 = 1.5 A$. (iv)

So, the current in bulbs B_1 and B_3 is $1.5 A$ and the ammeter A_1 read $1.5 A$ and A_3 read $1.5 A$ as well.

(iii) Power = Current $\times$ Potential difference.

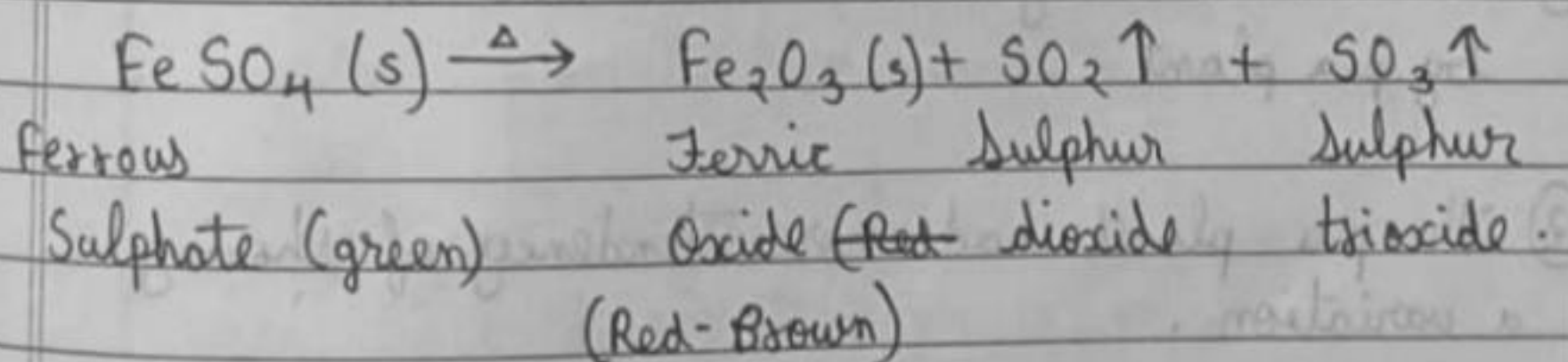
$P = IV$

For the entire circuit when all the bulbs are glowing,

$P = 3 \times 4.5$

$P = 13.5 W$.

Ans 36: (a) (i) The reaction is as follows:



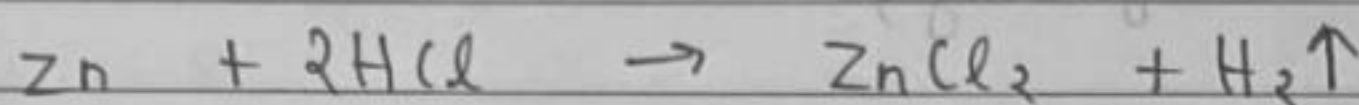
(ii) The colourless gas X is Sulphur dioxide and it has a smell of burning sulphur or burning matches.

(iii) This reaction is a decomposition reaction and is endothermic in nature.

(b) Zinc liberates Hydrogen gas when it reacts with Hydrochloric acid while copper does not because zinc reacts with Hydrochloric acid while copper does not.

Zinc is more reactive than hydrogen in reactivity series while copper is less reactive than hydrogen.

Due to this Zinc can displace hydrogen from HCl while copper cannot.



Ans 37: (i) Mendel selected a pea plant for his experiment because:

- ① There were many traits that were shown by pea plant.
- ② The pea plant had more tendency of showing a variation.
- ③ It is easy to grow pea and they take less time to grow.
- ④ The pea plants are bisexual, can be self pollinated or cross pollinated easily.

and are available in large number due to which Mendel choose it.

(ii) Dominant traits are those traits which show up even in ~~an~~ heterozygous conditions and are shown in F_1 progeny.

Recessive traits are those traits which show up only in ~~the~~ homozygous conditions and are shown directly in F_2 progeny and do not appear in F_1 progeny.

(iii) Seed shape : Dominant : Round
Recessive : wrinkled.

Seed colour : Dominant : ~~Green~~ Yellow
Recessive : Yellow ~~Green~~

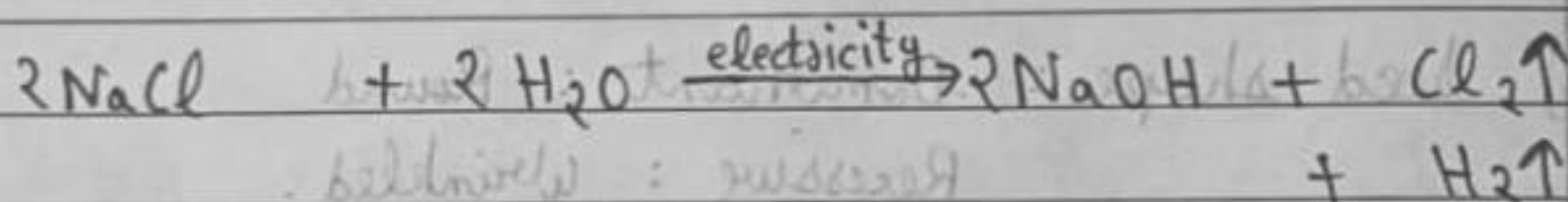
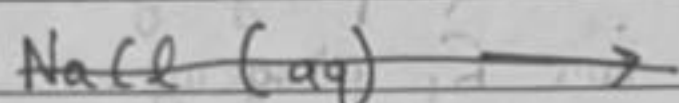
Ans 38. (a) The range of visible light spectrum is from 380 nm to 780 nm.

(b) Different wavelengths of energy is produced when light passes through a prism because a prism divides the white light coming from the light source into its constituent colours (VIBGYOR). All of these colours have different wavelengths of energy.

(c) The different colours travel at different speeds due to which they make different angles with the prism and hence dispersion of white light takes place.

Ans 39. (i) The chlor Alkali process involves the passing of electricity through brine or aqueous solution of sodium chloride.

It is as follows:



The products formed are Sodium Hydroxide, chlorine gas and hydrogen gas.

(ii) Chlorine gas is formed at Anode and Hydrogen gas is formed at Cathode.

(iii) Use of Chlorine:

(i) Used for purifying water and used in swimming pools, making Bleaching Powder etc.

Uses of Hydrogen is that it helps in making Ammonia, margarine and as fuel.

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